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MILITARY SPECIFICATION

# PLATES, MOUNTING, SAFETY AND ARMING DEVICE, XM-2, XM-3, AND M2E2

## 1. SCOPE

1.1 Scope. This specification covers three types of mounting plates for use with a safety and arming device.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-10025 — Packing and Marking for Domestic Shipping of Inert Ammunition Components

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-130 — Identification Marking of U. S. Military Property

MIL-STD-202 — Test Methods for Electronic and Electrical Equipment.

MIL-STD-354 — Temperature and Humidity Test for Use in Production of Fuzes

DRAWINGS

ORDNANCE CORPS

8030821 — Plate, Mounting, Safety and Arming Device, M2E2

8532426 — Plate, Mounting, Safety and Arming Device, XM2

8532476 — Plate, Mounting, Safety and Arming Device, XM3

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Preproduction sample. When a preproduction sample is required (see 6.2), it shall meet all requirements of this specification

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and shall be manufactured using the same methods proposed for the manufacture of subsequent production lots.

3.2 **Material.** The material used in the manufacture of the mounting plates shall be in accordance with Drawings 8030821, 8532426, and 8532476 and applicable drawings and specifications referenced thereon.

3.3 **Design and construction.** The design and construction of the mounting plates shall be in accordance with Drawings 8030821, 8532426, and 8532476 and applicable drawings and specifications referenced thereon.

3.4 **Electrical.**

3.4.1 *Ground potential.* The design and construction of the mounting plates shall insure that all external metallic parts are at ground potential at all times.

3.4.2 *Dielectric strength.* There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength, when the insulated areas of the mounting plates are subjected to an alternating current (ac), root mean square (rms) potential of 500 volts plus twice the rated voltage at a frequency of 60 cycles per second (cps) for a period of four (4) seconds, plus or minus one (1) second.

3.4.3 *Insulation resistance.* The insulation resistance between pins of the connector and the mounting plate or between the pin connectors and the mounting plate shall have a minimum resistance of 100 megohms when measured with 100 volts direct current (dc) applied.

3.4.4 *Circuitry resistance.*

3.4.4.1 The circuitry resistance between pin A of the connector and pin connector 2 or between pin C of the connector and pin connector 4 shall be 1000 ohms, plus or minus 100 ohms.

3.4.4.2 The circuitry resistance between pin connectors 1 and 2 or pin connectors 3 and 4 shall be 35.0 plus or minus 7 megohms with 100 volts dc applied.

3.4.5 *Circuit attenuation.* The circuit attenuation of the mounting plates shall be as specified in table I when tested as specified in 4.4.1.5.

TABLE I.

| Input | | | | Output | | |
|---|---|---|---|---|---|---|
| Pin connector | Pins | Volts | Frequency | Pin connector | Pins | Range volts |
| 1-2 | ..... | 0.1 | 25,000 cps | ..... | A-B | 0.043 - 0.062 |
| | A-B | " | " | 1-2 | ..... | " " |
| 3-4 | ..... | " | " | ..... | C-D | " " |
| | C-D | " | " | 3-4 | ..... | " " |

3.5 **Environmental.**

3.5.1 *Low temperature.* The mounting plates shall be capable of meeting the requirements for insulation resistance (see 3.4.3), circuitry resistance (see 3.4.4), and circuit attenuation (see 3.4.5), after subjection to a low temperature of minus 54 degrees centigrade (C), plus or minus 2.8 degrees C (minus 65 degrees Fahrenheit (F), plus or minus 5.0 degrees F) for a minimum period of four (4) hours. There shall be no cracks or breaks of the potting compound.

3.5.2 *High temperature.* The mounting plates shall be capable of meeting the requirements for insulation resistance (see 3.4.3), circuitry resistance (see 3.4.4), and circuit attenuation (see 3.4.5), after sub-



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jection to a high temperature of plus 74 degrees C, plus or minus 2.8 degrees C (plus 165 degrees F, plus or minus 5.0 degrees F) for a minimum period of four (4) hours There shall be no cracks or breaks of the potting compound.

3.5.3 *Vibration.* The mounting plates shall be capable of meeting the requirements for insulation resistance (see 3.4.3), circuitry resistance (see 3.4.4), and circuit attenuation (see 3.4.5), after subjection to vibrations at any frequency from 10 to 55 cps, inclusive, at an amplitude of 0.03 inch (0.06 inch total excursion).

3.5.4 *Temperature and humidity.* The mounting plates shall be capable of meeting the requirements for insulation resistance (see 3.4.3), circuitry resistance (see 3.4.4), and circuit attenuation (see 3.4.5), after subjection to the conditions specified in Standard MIL-STD-354.

**3.6 Details of components.** The components of the mounting plates shall be in accordance with Drawings 8030821, 8532426, 8532476, and applicable drawings and specifications referenced thereon.

**3.7 Dimensions.** Dimensions and tolerances of the mounting plates shall be in accordance with Drawings 8030821, 8532426, 8532476, and applicable drawings and specifications referenced thereon.

**3.8 Color and finish.** The color and finish of the mounting plates shall be in accordance with Drawings 8030821, 8532426, 8532476, and applicable drawings and specifications referenced thereon.

**3.9 Product marking.** Product markings shall be in accordance with Drawings 8030821, 8532426, 8532476, and applicable drawings referenced thereon and Standard MIL-STD-130.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

**4.2 Preproduction sample.** The preproduction sample shall be subjected to all examinations and tests specified herein. Unless otherwise specified (see 6.2), the Government will perform the examinations and tests at the contractors plant. If the preproduction sample does not meet all requirements of this specification, it will be rejected and returned to the contractor. Subsequent units or lots as applicable will not be considered for acceptance until approval of the preproduction sample has been obtained .

**4.3 Inspection.**

4.3.1 *Lot Size.* Lot size shall not exceed that quantity of mounting plates produced in one continuous operation by the same manufacturer and manufactured in accordance with the same drawing, drawing revision.

4.3.1.1 *Control sample.* Mounting plate assemblies selected at random from each accepted lot shall be shipped within two days after acceptance to the technical laboratory (see 6.2) for evaluation. However, at the direction of the contracting officer upon the request of the technical laboratory, the sample may be selected at any point on the production line during the production run. The results of the evaluation will not affect lot acceptance, but may form a basis for corrective action by mandatory changes in subsequent lots. The number of units in the control sample shall be as follows:



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| *Lot size* | *Control sample size* |
|---|---|
| up to 300 | 5 |
| 301 to 600 | 10 |
| 601 to 1,000 | 15 |
| over 1,001 | 20 |

The control sample may be reduced, or the sample discontinued at the discretion of the technical laboratory upon approval by the contracting officer.

4.3.2 *Sampling.*

4.3.2.1 *Inspection.* Unless otherwise specified (see 6.2.), sampling for inspection shall be in accordance with Standard MIL-STD-105, Inspection Level II and the classification of defects specified in table II.

*Table II. Classification of Defects in accordance with Standard MIL-STD-105.*

Mounting plate assembly (Drawing 8030821).

| Categories | Defects | Inspection method |
|---|---|---|
| Critical | AQL 0.015 percent | |
| 1. | Depth of primacord hole | Gage |
| 2. | Location of pin connectors, locating of pin holes and of primacord holes | Gage |
| Major | AQL 0.25 percent | |
| 101. | Mounting bracket holes (mounting plate to missile) | Gage |
| 102. | Pitch diameter of threads (internal threads) | Gage |
| 103. | Minor diameter of threads (internal threads) | Gage |
| 104. | Connector mounted tight and flush with shoulder | Visual and Manual |
| 105. | Connector pins or threads damaged | Visual |
| 106. | Pin connector damaged | Visual |
| 107. | Potting cracked or broken | Visual |
| 108. | Bubbles in potting compound | Visual |
| 109. | Nomenclature misleading or unidentifiable | Visual |
| 110. | Mate with female connector | Manual |

Mounting plate assembly (Drawing 8532426)

| Categories | Defects | Inspection method |
|---|---|---|
| Critical | AQL 0.015 percent | |
| 3. | Depth of primacord hole | Gage |
| 4. | Location of pin connectors, locating pin holes and of primacord hole | Gage |
| Major | AQL 0.25 percent | |
| 111. | Mounting bracket holes (mounting plate to missile) | Gage |
| 112. | Pitch diameter of threads (internal threads) | Gage |
| 113. | Minor diameter of threads (internal threads) | Gage |
| 114. | Diaphragm thickness | Gage |
| 115. | Connector mounted tight and flush with shoulder | Visual and Manual |
| 116. | Connector pins or threads damaged | Visual |
| 117. | Pin connector damaged | Visual |
| 118. | Potting cracked or broken | Visual |
| 119. | Bubbles in potting compound | Visual |
| 120. | Nomenclature misleading or unidentifiable | Visual |
| 121. | Mate with female connector | Manual |



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Mounting plate assembly (Drawing 8532476)

| Categories | Defects | Inspection method |
|---|---|---|
| Critical | AQL 0.015 percent | |
| 5. | Depth of primacord hole | Gage |
| 6. | Location of pin connector, locating pin holes and of primacord hole | Gage |
| Major | AQL 0.25 percent | |
| 122. | Mounting bracket holes (mounting plate to missile) | Gage |
| 123. | Pitch diameter of threads (internal threads) | Gage |
| 124. | Minor diameter of threads (internal threads) | Gage |
| 125. | Connector mounted tight and flush with casting | Visual and Manual |
| 126. | Connectors pins or threads damaged | Visual |
| 127. | Pin connectors damaged | Visual |
| 128. | Potting cracked or broken | Visual |
| 129. | Bubbles in potting compound | Visual |
| 130. | Nomenclature misleading or unidentifiable | Visual |

Preparation for delivery (inside package)

| Categories | Defects | Inspection method |
|---|---|---|
| Critical | | |
| None. | | |
| Major | AQL 0.65 percent | |
| 131. | Screws and/or lock washer missing | Visual |
| 132. | Dust cap missing | Visual |
| 133. | Contents loose | Manual and Audible |
| 134. | Damaged lower case container | Visual |
| 135. | Marking improper, misleading or unidentifiable | Visual |
| 136. | Seal missing | Visual |

Prime Package

| Categories | Defects | Inspection method |
|---|---|---|
| Critical | | |
| None. | | |
| Major | AQL 0.5 percent | |
| 137. | Marking improper, misleading or unidentifiable | Visual |
| 138. | Improper lower case seal | Visual |
| 139. | Contents loose or exposed | Visual |
| 140. | Container damaged | Visual |

1.3.2.2 *For acceptance testing.* Acceptance testing shall be performed on 100 percent of production and with the following characteristics in accordance with their corresponding test paragraphs.

| *Characteristics* | *Paragraphs* |
|---|---|
| Insulation resistance | 4.4.1.3 |
| Circuitry resistance | 4.4.1.4 |
| Circuit attenuation | 4.4.1.5 |

4.3.2.3 *For inspection testing.* Inspection testing shall be performed on the preproduction and control samples only and with the following characteristics in accordance with their corresponding test paragraphs.

| *Characteristics* | *Paragraphs* |
|---|---|
| Low temperature | 4.5.1 |
| High temperature | 4.5.2 |
| Vibration | 4.5.3 |
| Temperature and humidity | 4.5.4 |

**4.4 Test procedures.** The following tests shall be performed on 100 percent of production with the exception of the environmental tests which shall be performed on the preproduction sample only. The preproduc-



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tion sample shall be tested in the sequence specified.

4.4.1 *Electrical.*

4.4.1.1 *Ground potential.* All external metallic parts shall be measured to ground, using an ohm-meter (2 percent accuracy) to determine conformance to the requirements specified in 3.4.1.

4.4.1.2 *Dielectric strength.* The mounting plates shall be tested in accordance with Standard MIL-STD-202, method 301, to determine conformance to the requirements specified in 3.4.2.

4.4.1.3 *Insulation resistance.* Immediately after satisfactorily withstanding the test for dielectric strength the mounting plates shall be given the insulation resistance test in accordance with Standard MIL-STD-202, method 302, condition A, to determine conformance to the requirements specified in 3.4.3.

4.4.1.4 *Circuitry resistance.*

4.4.1.4.1 A circuitry resistance test of the mounting plates shall be conducted using an ohm-meter (2 percent accuracy) to determine conformance to the requirements specified in 3.4.4.1.

4.4.1.4.2 A circuitry resistance test of the mounting plates shall be conducted using a megohmmeter (3 percent accuracy) to determine conformance to the requirements specified in 3.4.4.2.

4.4.1.5 *Circuit attenuation.* Connect a 4.7 kilohm fixed resistor in series with a 250 kilohm variable resistor between pin connector 2 and pin A of the connector and adjust the resistor until the resistance between the terminals is 900 ohms, plus or minus 5 percent. Using a signal source having an output impedance of not more than 42 ohms, apply inputs as specified in table I. Measure the output voltage of pins A and B and pin connectors 1 and 2, respectively using a vacuum tube voltmeter to determine conformance to the requirements specified in 3.4.5. Repeat the test except that pin connectors 3 and 4 shall be substituted for pins 1 and 2, respectively, and pins C and D of the connector shall be substituted for pins A and B, respectively.

4.5 **Environmental.**

4.5.1 *Low temperature.* The mounting plates shall be placed in an environmental test chamber and subjected to a low temperature of minus 54 degress C plus or minus 2.8 degrees C (minus 65 degrees F plus or minus 5.0 degrees F) for a minimum period of four (4) hours. The mounting plates shall be tested for insulation resistance (see 4.4.1.3), circuitry resistance (see 4.4.1.4), and circuit attenuation (see 4.4.1.5) within two (2) minutes after being removed from the environmental test chamber to determine conformance to the requirements specified in 3.5.1.

4.5.2 *High temperature.* The mounting plates shall be placed in an environmental test chamber and subjected to a high temperature of plus 74 degrees C plus or minus 2.8 degrees C (plus 165 degrees F plus or minus 5.0 degrees F) for a minimum period of four (4) hours. The mounting plates shall be tested for insulation resistance (see 4.4.1.3), circuitry resistance (see 4.4.1.4), and circuit attenuation (see 4.4.1.5), within two (2) minutes after being removed from the environmental test chamber to determine conformance to the requirements specified in 3.5.2.

4.5.3 *Vibration.* The mounting plates shall be tested as specified in method 201 of Standard MIL-STD-202. The mounting plates shall then be tested for insulation resistance (see 4.4.1.3), circuitry resistance (see 4.4.1.4), and circuit attenuation (see 4.4.1.5) to determine conformance to the requirements specified in 3.5.3.

4.5.4 *Temperature and humidity.* The



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mounting plates shall be tested as specified in Standard MIL-STD-354. The mounting plates shall then be tested for insulation resistance (see 4.4.1.3), circuitry resistance (see 4.4.1.4), and circuit attenuation (see 4.4.1.5) to determine conformance to the requirements specified in 3.5.4.

**4.6 Preservation, packaging, packing, and marking.** The preservation, packaging, packing and marking shall be inspected in accordance with specification MIL-P-10025 with the following exceptions:

(a) Each inside package shall contain one (1) plate with plastic dust cap in place, one (1) bag containing cross-recessed screws, ten (10) each when shipping XM-2 or M2E2, four (4) each when shipping XM-3 and an equal number of lock washers, and sufficient dunnage to prevent movement of plate in any direction.

(b) Inside package shall be sealed with ¾" pressure sensitive tape on both ends and both sides.

## 5. PREPARATION FOR DELIVERY

**5.1 Preservation, packaging, packing and marking.** Preservation, packaging, packing and marking for the required level (see 6.2) shall be in accordance with the applicable packaging data sheet for Ordnance Part Number 8030821, 8532426, and 8532476.

## 6. NOTES

**6.1 Intended use.** The mounting plate assemblies are a combination mounting fixture and low band filter for the M30A1 Safety and Arming Device.

**6.2 Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Part number or part numbers of mounting plates required.

(c) Whether a preproduction sample is required (see 3.1).

(d) Where a preproduction sample test shall be performed (see 4.2.).

(e) Inspection other than specified (see 4.3).

(f) Facility to which preproduction and control samples are to be shipped (see 4.3.1.1).

(g) Selection of applicable levels of preservation, packaging packing, and marking (see 5.1).

(h) Any special marking required (see 5.1).

**Custodian:**
**Army—(MI)**

**Preparing activity:**
**Army—(MI)**

**Project No. 1420-A045**



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